7 July 1975
Superseding
MIL-T-58060A
29 June 1974

MILITARY SPECIFICATION

TABLE ASSEMBLY, SCORSBY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a roll-pitch-yaw table identified as a Scorsby Table Assembly.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATION

FEDERAL

CC-M-636	Motors, fractional horsepower (alternating current).
QQ-C-320	Chromium plating (electrodeposited).
QQ-N-290	Nickel plating (electrodeposited).
QQ-P-416	Plating, cadmium (electrodeposited).

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QQ-Z-325	Zinc coating, electrodeposited, Requirements for.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-636	Boxes, Shipping, Fiberboard.

MILITARY

MIL-M-3171	Magnesium Alloy, processes for pretreatment and prevention of corrosion on.
MIL-S-5002	Surface treatments and inorganic coatings for metal surfaces of weapons systems.
MIL-C-5541	Chemical conversion coating on aluminum and aluminum alloys.
MIL-S-7742	Screw threads, standard, optimum selected series: General specification for.
MIL-S-8512	Support equipment, aeronautical, special, General specification for the design of.
MIL-A-8625	Anodic coatings, for aluminum and aluminum alloys.
MIL-E-16298	Electric machines having rotating parts and associated repair parts; packaging of.

STANDARDS

FEDERAL

FED-STD-595	Color.

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MIL-STD-129	Marking for shipment and storage.
MIL-STD-130	Identification marking of U. S. military property.
MIL-STD-143	Standards and specification, order of precedence for selection of
MIL-STD-454	Standard general requirements for electronic equipment.
MIL-STD-808	Finish, protective, and codes for finishing schemes for ground and ground support equipment.
MIL-STD-810	Environmental test method.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Design.--The Scorsby Table Assembly shall be designed to shop-test aircraft gyroscopic instruments. It shall consist of the following major components: Electrically driven table which simulates the combination of roll-pitch-yaw motions of aircraft in flight; tilting head which supports the table; housing which shrouds the motor and reduction gearing; adjustable base; and adapter(s) or universal mount(s). The equipment design shall consider the safety requirements of MIL-STD-454 Requirement 1.

3.1.1 First Article.--If first article approval and functional tests are required (see 6.2), approval and testing shall be in accordance with 4.2.1. Approval of the first article shall not relieve the contractor of his responsibility to furnish a product in accordance with this specification. Any change or deviation from the approved first article shall be subject to the prior written approval of the contracting officer.

3.2 Materials.--Materials shall be as specified in MIL-S-8512. When specifications and standards are not specifically designated, selection of materials and processes shall be in accordance with MIL-STD-143.

3.3 Components.--General arrangement of components shall be as shown on figure 1.

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3.3.1 Base.--The base shall be rigid to the unit during all phases of the table assembly operation. It shall be capable, by hand adjustment, of leveling the entire unit. The base contact points shall not mar the surface of the bench on which it will operate.

3.3.2 Housing.--The housing enclosing the motor and reduction gearing may serve as the motor mounting surface when applicable. The housing shall provide suitable protection to prevent contact with moving parts.

3.3.3 Motor.--The table assembly shall be powered by a totally enclosed, synchronous, 110-volt, 60-cycle alternating current (ac) motor conforming to Specification CC-M-636, excepting speed which shall be in accordance with the performance specified in 3.5.

3.3.4 Tilting Head.--A tilting head shall be incorporated in the table assembly to enable the table assembly to tilt from a horizontal plane, plus or minus 15 degree - plus or minus 2 degrees. The included angle of travel is 30 degrees. A tilt scale shall indicate the degrees of tilt to with plus or minus 1 degree. The scale shall be graduated in 1-degree increments with numerical designations every 5 degrees, plus a mark at 1-1/2 degrees and at 7-1/2 degrees.

3.3.5 Table.--A table shall be attached to the tilting head (see fig. 1) to serve as a base for the installation of adapter(s) or universal mount(s) (see 3.3.8).

3.3.6 Level.--A removable spirit level shall be attached to the table, and positioned in a manner that when the table is in a horizontal position (zero degree on the tilt scale (see 3.3.4)), the bubble will center in the ring glass. The level shall be readily visible when the instruments to be tested are mounted, and its location shall not interfere with table motion or attachment of the instrument.

3.3.7 Locking.--A positive locking device which is easily accessible without disturbing the level position of the base, shall be provided to lock the position of the table at any degree of tilt over the range specified in 3.3.4.

3.3.8 Adapter(s) or universal mount(s).--If specified (see 6.2), adapter(s) or universal mount(s) shall be provided as mounting bracket(s) for the gyroscopic instruments to be tested.

3.4 Construction.--The Scorsby table shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installations, and service, without any parts working loose.

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Disassembly, service maintenance, and reassembly shall be possible with normally available commercial tools. Table assembly with rated load capacity of 25 pounds shall be capable of performing all requirements specified in 3.5 through 3.5.4.

3.4.1 Weight.--Weight of the Scorsby Table Assembly, ready for operation, excluding the adapter(s) or mount(s) (see 3.3.8), shall not exceed 35 pounds.

3.4.1.2 Dimensions.--Table assembly shall not be more than 12 inches wide, 12 inches deep, and 11 inches high without the adapter(s) or mount(s) (see 3.3.8). The table surface (see 3.3.5) shall be 10 to 12 inches square, and shall have four pairs of 10-32 tapped holes centrally located 1, 2, 3, and 4 inches from the front edge and 3 inches between centers.

3.4.2 Standard parts.--Standard parts (AN or MS) shall be used wherever they are suitable for the purpose, and shall be identified by their part numbers. Commercial utility parts such as screws, bolts, nuts, cotter pins, etc., may be used, provided they have suitable properties and are replaceable by AN or MS standard parts without alteration, and provided the corresponding AN or MS standard part numbers are referenced in the instruction data or in the parts listing (see 3.8). In applications for which no suitable corresponding AN or MS standard part is in effect on date of invitation for bids, commercial parts may be used provided they conform to all applicable requirements of this specification.

3.4.3 Screw threads.--Threaded parts shall conform to MIL-S-7742.

3.4.4 Interchangeability of parts.--All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance.

3.5 Performance.--The Scorsby table assembly shall be capable of testing instruments weighing up to 25 pounds (with load point at the center of gravity of plate and a cube compatible with the 12 inch table assembly). Instruments shall face the front of the table. The center of gravity of the instruments under test shall be approximately over the center of the table surface.

3.5.1 Scorsby Motion.--The table shall create the combined motions of roll-pitch-yaw. This combined motion is known as scorsby motion. During operation, with any angle between zero and 15 degrees set in the tilting head, the magnitude of the roll, pitch, and yaw oscillation shall be within 20 percent of the set in value.

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3.5.2 Reversing motion.--The assembly shall incorporate a reversing mechanism which allows for both automatic and manual reversing of the scorsby motion. The automatic mechanism shall reverse the scrosby motion every 6, ± 1/2, revolutions (cycles) and the manual control shall override the automatic reversing mechanism.

3.5.3 Rate of motion. The rate of scorsby motion shall be 6 motions, ± 1/2-motion per minute.

3.5.4 Exposure to extreme temperatures.--The table assembly shall be capable of withstanding, without adverse effects, temperatures of minus 32°C (-25°F) to plus 52°C (125°F) during shipping and storage.

3.5.4.1 High Temperature.--The table assembly shall be designed to be capable of operation in an ambient temperature of 52°C (125°F) ± 2°C.

3.5.4.2 Low Temperature.--The table assembly shall be designed to be capable of operation in an ambient temperature of minus 18°C (0°F) ± 2°C.

3.6 Identification.--Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.7 Color. The exterior color of the item shall be in accordance with MIL-STD-808, Color No. 16473 of FED-STD-595.

3.8 Plated Surfaces.--Metal surfaces not painted shall be protected by other suitable means such as plating and anodizing. Methods shall be selected in accordance with Specification MIL-S-5002. Surface finishes shall be in accordance with the applicable specifications listed in Table 1.

Table 1 Metal Surface Finishes

Surface Treatment	Specification
Cadmium	QQ-P-416
Zinc Plating	QQ-Z-325
Nickel Plating	QQ-N-290
Chromium Plating	QQ-C-320
Anodizing of Aluminum	MIL-A-8625
Chemical Treatment of Aluminum	MIL-C-5541
Protective Treatment of Magnesium	MIL-M-3171

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3.9 Workmanship.--The table assembly, including all parts and accessories shall be constructed and finished to produce a product free from defects which could effect proper functioning in service. Particular attention shall be given to neatness, thoroughness of soldering, wiring, brazing, painting, machine screw assemblies, and freedom of all parts from burrs and sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.--Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of examinations and tests.--Examinations and tests shall be classified as (1) first article, (2) quality conformance.

4.2.1 First Article.--The first Table Assembly shall be tested as specified in 4.3 (see 6.2). Data shall be recorded and used as the Pretest Performance Data for the Environmental test.

4.2.2 Quality conformance.--Each Scorsby Table Assembly shall be submitted to the test specified in 4.3.1 and 4.3.2.

4.3 Tests.--

4.3.1 Examination.--Each Scorsby Table Assembly shall be inspected to determine compliance with this specification.

4.3.2 Performance.--Each Table Assembly shall be subjected to the performance tests.

4.3.2.1 Scorsby Motion Test.--The Table Assembly shall be tested to verify the roll-pitch-yaw or Scorsby motion requirement. During operation, with any angle between zero and 15 degrees set in the tilting head, the magnitude of the roll, pitch, and yaw oscillation shall be within 20 percent of the set in value.

4.3.2.2 Reversing Motion Test.--The Table Assembly shall be tested to verify the reversing motion requirement. The automatic mechanism shall reverse the Scorsby motion every 6, plus or minus 1/2, revolutions (cycles) and the manual control shall override the automatic reversing mechanism.

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4.3.2.3 Rate of Motion Test.--The rate of motion shall be tested to verify the rate of 6 motions, plus or minus 1/2-motion per minute.

4.3.3 Environmental.--The first Scorsby Table Assembly shall be subjected to the following tests.

4.3.3.1 High Temperature.--The high temperature test shall be conducted in accordance with MIL-STD-810, Method 501, Procedure II except that the maximum temperature shall not exceed 52°C (125°F) + 2°C. The length of time required for operation and measurement shall not exceed one hour. The equipment shall meet the test requirements of 4.3.2.1, 4.3.2.2, and 4.3.2.3.

4.3.3.2 Low Temperature.--The low temperature test shall be conducted in accordance with MIL-STD-810, Method 502, Procedure I. The storage temperature shall be -32°C (-25°F) + 2°C for 72 hours and then be stabilized at -18°C (0°F) + 2°C. The equipment shall be operated at -18°C, + 2°C, and shall meet the test requirements of 4.3.2.1, 4.3.2.2, and 4.3.2.3.

4.4 Inspection of preparation for delivery.--Preservation, packaging, packing, and marking shall be inspected to assure compliance with section 5 of this specification.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.--Each Scorsby Table shall be preserved and packaged Level A or C, as specified (see 6.2) in accordance with MIL-E-16298.

5.2 Packing.--Scorsby Tables, preserved and packaged per 5.1, shall be packed Level A, B, or C as specified (see 6.2) in accordance with MIL-E-16298 and as follows.

5.2.1 Level A.--Shipping containers shall conform to PPP-B-601, overseas type.

5.2.2 Level B.--Shipping containers shall conform to V2s of PPP-B-636 or domestic type of PPP-B-601.

5.3 Marking.--In addition to any special marking required by the contract or order, marking shall be in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use.--The Scorsby Table Assembly is to be bench-mounted and used for determining the serviceability of gyroscopic instruments used on aircraft.

6.2 Ordering data.--Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Whether first article approval and/or functional tests are required (see 3.1.1).

(c) Number and type of adapters or mounts required (see 3.3.8).

(d) Level of preparation for delivery required (see 5.1 and 5.2).

(e) Whether special marking is required (see 5.3).

6.3 Amendment 1 dated 26 November 1968 to MIL-T-58060A is included in this revision.

Custodians:

Army - AV
Navy - AS
Air Force - 82

Review Interest:

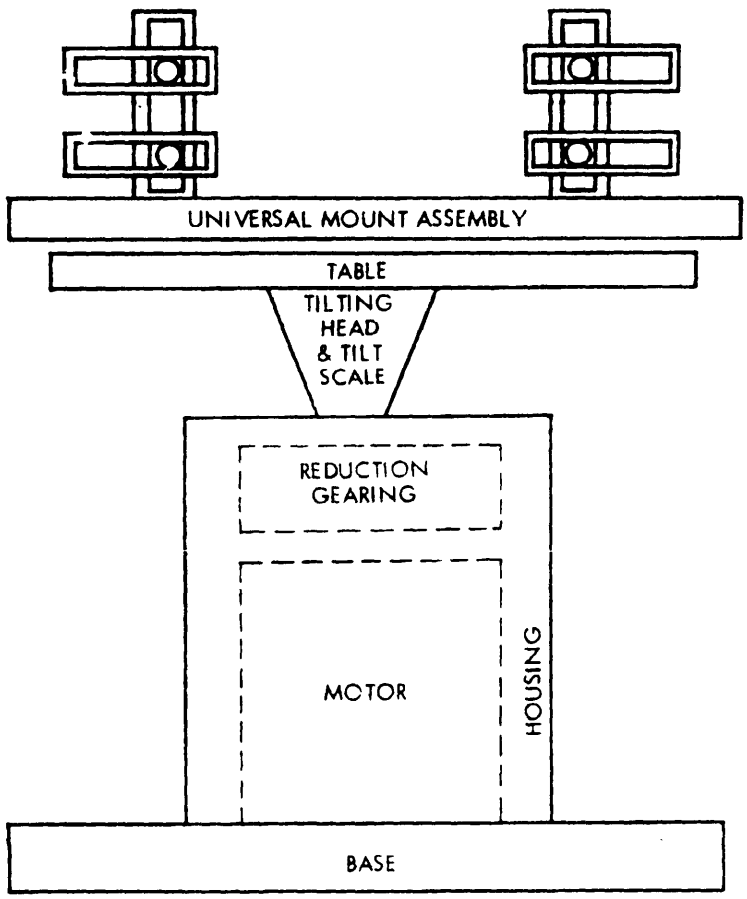
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SCORSBY TABLE ASSEMBLY, GENERAL ARRANGEMENT
FIGURE 1

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CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

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A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

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